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Beyond Data Manipulation

Week 4 - Assignment

Programming Assignment: Assignment 4 3h

Quiz: Final Quiz Submitted

Reading: Post-course Survey 10 min

Reading: Keep Learning with Michigan Online! 10 min

Reading: Progress note from Christopher Brooks 3 min

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1. Consider the given NumPy arrays **a** and **b**. What will be the value of **c** after the following code is executed?

1 / 1 point

Try again

Receive grade

To Pass 80% or higher

Your grade 90%

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```
1 import numpy as np
2
3 a = np.arange(8)
4 b = a[4:6]
5 b[:] = 40
6 c = a[4] + a[6]
```

Like

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46

Correct

Since array slices are passed by reference, the values at indices 4 and 5 in **a** will also be modified after the values in the array slice **b** are changed to 40. Hence, after the code is run, **a** = array([0, 1, 2, 3, 40, 40, 6, 7]). As **a[4]** = 40 and **a[6]** = 6, **c** = 40 + 6 = 46.

2. Given the string **s** as shown below, which of the following expressions will be **True**?

1 / 1 point

```
2 s = 'ABCAC'
```

☐

```
1 len(re.search('A', s)) == 2
```

☒

```
1 bool(re.match('A', s)) == True
```

☐

```
1 re.match('A', s) == True
2
```

☐

```
1 len(re.split('A', s)) == 2
```

Correct

re.match('A', s) return an re.match object. Since it will find a match for 'A' in 's', thus converting this object to bool will return True.

3. Consider a string **s**. We want to find all characters (other than A) which are followed by triple A, i.e., have AAA to the right. We don't want to include the triple A in the output and just want the character immediately preceding AAA. Complete the code given below that would output the required result.

1 / 1 point

```
1 def result():
2     s = 'ACAABAACAABACBADD0FSD0DFSSSSADAFAAACBAAAFASD'
3
4     result = []
5     # complete the pattern below
6     pattern = '([A])AAA'
7     for item in re.finditer(pattern, s):
8         # identify the group number below.
9         result.append(item.group(1))
10
11     return result
```

Run

Reset

No Output

Correct

Good job!

4.

1 / 1 point

```
>>> df
d      4
b      7
a     -5
c      3
dtype: int64
```

Consider the following 4 expressions regarding the above pandas Series **df**. All of them have the same value except one expression. Can you identify which one it is?

☐ 1 df.iloc[0]

☐ 1 df['d']

☐ 1 df[0]

☒ 1 df.index[0]

Correct

df.index[0] = 'd', which is different from the other three which are equal to 4

5.

1 / 1 point

```
>>> s1
Mango      20
Strawberry 15
Blueberry  18
Vanilla     31
dtype: int64

>>> s2
Strawberry  20
Vanilla     30
Banana      15
Mango       20
Plain       20
dtype: int64
```

Consider the two pandas Series objects shown above, representing the no. of items of different yogurt flavors that were sold in a day from two different stores, **s1** and **s2**. Which of the following statements is **True** regarding the Series **s3** defined below?

```
1 s3 = s1.add(s2)
```